

Continuous Pad-Steam & Thermosol Dyeing Machinery Constraints Manual

1. Overview of Continuous Wet Processing Constraints

In continuous dyeing (Pad-Steam, CPB, Thermosol), fabric moves at high speeds through padding mangles, pre-dryers, fixation chambers, and wash boxes.

2. Target Wet Pick-up Rates (E%)

Substrate	Target Wet Pick-up (E%)	Nip Pressure Range	Roller Hardness
100% Cotton Woven	65% - 70%	2.5 - 4.0 bar	70 - 75 Shore A
100% Polyester Woven	35% - 45%	3.5 - 5.0 bar	75 - 80 Shore A
P/C 65/35 Blend	55% - 60%	3.0 - 4.5 bar	70 - 75 Shore A
100% Viscose / Rayon	75% - 85%	2.0 - 3.5 bar	65 - 70 Shore A

3. Fixation Chamber Machine Constraints

Steamer Chamber: Saturated steam 102°C - 104°C, Oxygen < 0.1% for Vat reduction, Dwell 45 - 90 s.

Thermosol Unit: Temp 200°C - 215°C (± 1.5°C across width), Dwell 45 - 60 s, IR Pre-drying to 25% moisture.